TrES-2b

R-0045

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_250906` · created 2026-07-09T04:45:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460924.7671 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.102 +/- 0.044
Transit depth (Rp/Rs)^2	1.03 +/- 0.88 [%]
Orbital Inclination (inc)	83.67 +/- 2.94
Airmass coefficient 1 (a1)	1.84 +/- 1.12
Airmass coefficient 2 (a2)	-0.55 +/- 0.55
Scatter in the residuals of the lightcurve fit is	56.94 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	3.5
Transit Duration (day)	0.056 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.102 ± 0.044 vs 0.1254 (deviation 0.49σ)
Duration fit vs published	0.056 d vs 0.0746 d (deviation 0.42σ)
Mid-time O–C	2.7 min
Depth SNR	2.1
Residual scatter	56.94 %

Light curve

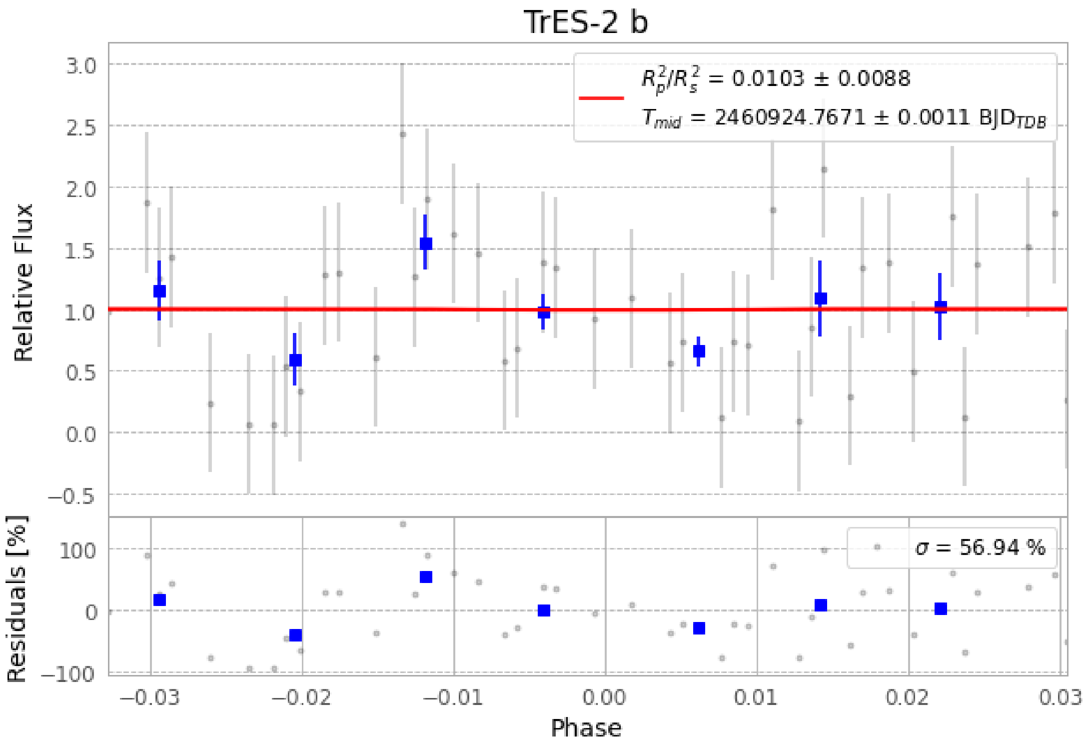

Figure 1 — FinalLightCurve_TrES-2 b_2025-09-05.png (SHA-256 0fde88739c065944…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 282], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 11eea4da5f0038ba…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

77 FITS frames (51 MB), `TRES-2250906042416.FITS` → `TRES-2250906081215.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-250906.log (5870e529045d…), FinalLightCurve_TrES-2 b_2025-09-05.png (0fde88739c06…), FinalLightCurve_TrES-2 b_2025-09-05.csv (a8e02521e689…)

Compute resources

Wall time 230.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)